

Technology Stack (Architecture & Stack)

Date: 31 January 2025 **Team ID:** SWTID-2026-7464 **Project Name:** Script-Controlled ACL – Restrict Record Access Based on Field Value **Maximum Marks:** 4 Marks

Technical Architecture:

The proposed architecture treats the ACL script as the application-security decision point. A user requests a record; the platform evaluates the configured ACL script against the record's field value and applicable user context; the data store returns the record only when access is permitted. The stack below is a proposed implementation mapping for this project.

User / Client		Application / Access-Control Layer		Data Layer
User requests a record	→	ServiceNow ACL Script evaluates field value + user context	→	ServiceNow Record Data Table / Record Storage
	←	Allow / Deny decision	←	Record + field values

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	User interacts with the protected application and requests records.	ServiceNow Web UI / Browser
2	Application Logic-1	ACL script evaluates the target record field value and access context.	ServiceNow ACL + JavaScript
3	Application Logic-2	Access decision logic returns allow or deny for the requested record.	ServiceNow Access Control Rules / Script
4	Application Logic-3	Administration and testing of field-based access conditions.	ServiceNow ACL Configuration / Admin Console
5	Database	Stores protected records, fields, and configuration data.	ServiceNow Platform Database / Tables
6	Cloud Database	Platform-managed database service supporting the application data layer.	ServiceNow-managed Cloud Database
7	File Storage	Optional storage for supporting documentation or test evidence.	ServiceNow Attachments / Platform Storage
8	External API-1	No external API is required for the core ACL decision.	Not required for core solution
9	External API-2	External integrations can be added only if a business requirement requires them.	Optional REST API / Integration
10	Machine Learning Model	No machine-learning model is required for deterministic field-value ACL evaluation.	Not applicable
11	Infrastructure (Server / Cloud)	Application deployment and execution environment.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core ACL implementation; the platform's scripting/configuration facilities are used.	JavaScript-based ServiceNow scripting
2	Security	Field-based ACL enforcement, authenticated user context, role/permission checks, and deny-by-default behavior for non-matching conditions.	ServiceNow ACLs, Roles, Authentication, HTTPS/TLS, Security Controls
3	Scalable Architecture	Centralized platform ACL logic can be reused across additional tables, fields, conditions, and user contexts.	ServiceNow Cloud Platform / Modular ACL Rules

S.No	Characteristics	Description	Technology
4	Availability	The solution runs within the managed platform environment, avoiding a separate custom authorization server for the core use case.	ServiceNow-managed Cloud Infrastructure
5	Performance	Keep ACL scripts concise, evaluate only required fields/context, and avoid unnecessary database or external calls during authorization.	Server-side JavaScript / ServiceNow ACL execution

Template alignment: The uploaded template asks for an architectural diagram, infrastructure demarcation, external interfaces, data storage, and (if applicable) ML interfaces, followed by Components & Technologies and Application Characteristics. This document follows those sections. The ServiceNow stack is presented as a proposed technology mapping because the template itself does not prescribe a platform.